

12 - WEEK ADVANCED TRACK

# DEVOPS & PLATFORM ENGINEERING

From Git fundamentals to GitOps, Kubernetes at scale, DevSecOps, observability and AI-assisted operations — built with industry mentors, six real-time projects and one production-grade capstone.

KUBERNETES

TERRAFORM

ARGO CD

GITHUB ACTIONS

DOCKER

ANSIBLE

PROMETHEUS

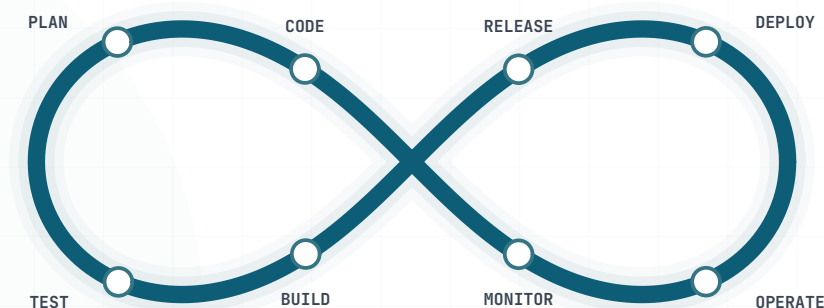
OPENTELEMETRY

VAULT

ISTIO

AWS

AZURE



**6 real-time projects • 1 capstone • Live mentorship • Placement support**

W H O   W E   A R E

# Learning engineered for the industry you will join.

Edufyi Tech Solutions transforms the educational journey for college students through cutting-edge virtual internships and structured, mentor-led training. Every learner ships real infrastructure, reviews real pull requests and defends a real capstone.

**12**WEEKS OF  
MENTORSHIP**06**REAL-TIME  
PROJECTS**25+**TOOLS IN THE  
STACK**01**CAPSTONE  
DEFENCE**MISSION**

To equip students with virtual internships and production-grade projects that convert into placements, guided end-to-end by working industry mentors.

**VISION**

To transform education by linking academic learning to real operational practice, so graduates arrive job-ready on day one.

W H O   I S   I T   F O R

## Built for four kinds of learners

- **Final-year & pre-final students**

CS / IT / ECE students who want an infrastructure-first career path.

- **Fresh graduates**

Bridge the gap between coursework and what hiring teams actually test.

- **Manual QA & support engineers**

Move laterally into CI/CD, automation and reliability roles.

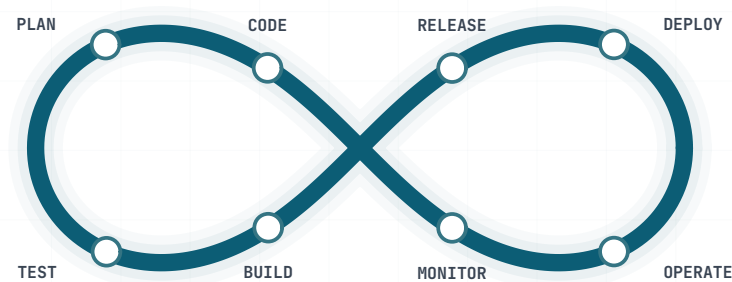
- **Junior developers**

Own your pipelines, containers, cloud footprint and on-call quality.

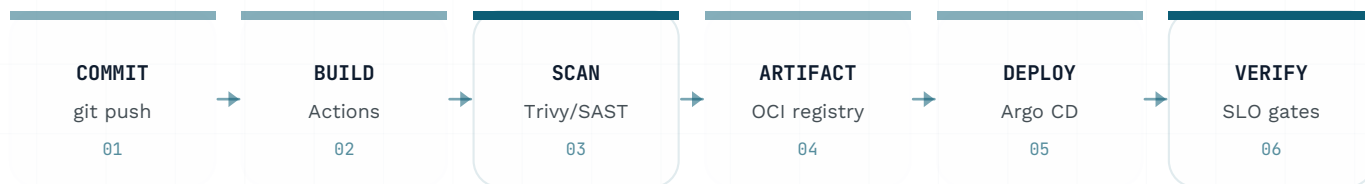
## HOW THE WORK FLOWS

# The lifecycle you will operate

Every module maps to a stage of the loop. You do not learn tools in isolation — you learn the hand-offs between them.



## REFERENCE DELIVERY PIPELINE



## THREE ENGINEERING PILLARS

### AUTOMATE

Everything reproducible: IaC, pipelines, policy as code.

### OBSERVE

Metrics, logs, traces and SLOs before incidents, not after.

### SECURE

Shift-left scanning, secrets management and supply-chain trust.

## HOW WE MEASURE YOU: DORA METRICS

### DEPLOY FREQUENCY

01

Ship on demand, many times a day.

### LEAD TIME FOR CHANGE

02

Commit to production in under an hour.

### CHANGE FAILURE RATE

03

Keep failed changes under 15%.

### TIME TO RESTORE

04

Recover service in minutes, not days.

## LESSON PLAN

# Weeks 01 – 03

WEEK  
**01**

## DevOps Foundations & SDLC

Culture, lifecycle and the operating model

- Why DevOps: flow, feedback, continuous learning
- SDLC models vs. DevOps operating model
- DORA four key metrics & elite performance
- Value-stream mapping a delivery pipeline
- Linux essentials, shell, systemd, networking

basics  
LINUX

BASH

DORA

AGILE

WEEK  
**02**

## Git, Trunk-Based Delivery & Code Review

Version control the way product teams use it

- Git internals: objects, refs, rebase vs. merge
- Trunk-based development & feature flags
- Git Flow, release branches, semantic versioning
- PR reviews, CODEOWNERS, conventional commits
- Pre-commit hooks, linting and signed commits

GIT

GITHUB

PRE-COMMIT

SEMVER

WEEK  
**03**

## Continuous Integration at Scale

Fast, trustworthy build feedback

- GitHub Actions: matrices, caching, reusable workflows
- Jenkins pipelines as code & shared libraries
- Unit, integration and contract test stages
- Build artifacts, caching and pipeline performance
- Quality gates with SonarQube coverage

thresholds  
GITHUB ACTIONS

JENKINS

GITLAB CI

SONARQUBE

## LESSON PLAN

# Weeks 04 – 06

WEEK  
**04**

## Containers & Image Supply Chain

From Dockerfile to hardened, signed images

- Container internals: namespaces, cgroups, OverlayFS
- Docker Compose for local multi-service stacks
- Vulnerability scanning, SBOM and Cosign signing
- Multi-stage builds, distroless & slim images
- Registries, tagging strategy, image promotion

DOCKER

BUILDKIT

TRIVY

SBOM

COSIGN

WEEK  
**05**

## Infrastructure as Code with Terraform

Cloud infrastructure that is reviewable

- Terraform core: providers, state, plan/apply
- Reusable landing zones for AWS / Azure
- Policy as code with Sentinel / OPA & Terratest
- Modules, workspaces, remote state & locking
- Drift detection, imports and safe destroys

TERRAFORM

AWS

AZURE

OPA

WEEK  
**06**

## Configuration Management & Cloud Ops

Repeatable servers, repeatable environments

- Ansible playbooks, roles, inventories, Vault
- Packer golden images & autoscaling groups
- Cost awareness and FinOps tagging discipline
- Idempotency and immutable infrastructure
- Core cloud services: IAM, VPC, S3, RDS, load balancers

ANSIBLE

PACKER

IAM

FINOPS

## LESSON PLAN

# Weeks 07 – 09

WEEK  
**07**

## Kubernetes Fundamentals

Orchestration you can reason about

- Architecture: control plane, kubelet, scheduler, etcd
- Storage classes, PVs, StatefulSets and jobs
- kubectl debugging and cluster troubleshooting
- Pods, Deployments, Services, Ingress, ConfigMaps
- Probes, resource requests/limits, HPA autoscaling

KUBERNETES

KUBECTL

EKS

AKS

WEEK  
**08**

## Advanced Kubernetes & Service Mesh

Operating clusters like a platform team

- Helm charts, Kustomize overlays, environment promotion
- Operators and Custom Resource Definitions
- Progressive delivery: canary and blue-green
- RBAC, network policies, admission control
- Istio / Linkerd: mTLS, traffic shifting, retries

HELM

KUSTOMIZE

ISTIO

ARGO ROLLOUTS

WEEK  
**09**

## GitOps & Continuous Delivery

Git as the single source of truth

- GitOps principles and reconciliation loops
- Environment promotion & release strategies
- Rollback, disaster recovery and change safety
- Argo CD / Flux: app-of-apps, sync waves, drift repair
- Secrets in GitOps: Sealed Secrets, External Secrets

ARGO CD

FLUX

GITOPS

SEALED SECRETS

## LESSON PLAN

# Weeks 10 – 12

WEEK  
10

## DevSecOps & Supply-Chain Security

Security as a pipeline stage, not a gate

- Threat modelling and shift-left security
- HashiCorp Vault: dynamic secrets and rotation
- SAST, DAST, SCA and secret scanning in CI
- OPA / Kyverno policy enforcement in clusters
- SLSA levels, provenance and compliance evidence

VAULT

TRIVY

KYVERNO

SLSA

ZAP

WEEK  
11

## Observability & Site Reliability

Know before your users tell you

- Metrics, logs, traces: the three signals
- Prometheus, Grafana, Alertmanager, Loki, Tempo
- OpenTelemetry instrumentation end-to-end
- SLI / SLO / error budgets and on-call practice
- Incident response, postmortems, chaos engineering

PROMETHEUS

GRAFANA

OPENTELEMETRY

LOKI

SRE

WEEK  
12

## Platform Engineering, AIOps & Capstone

Where DevOps is heading next

- Internal developer platforms & Backstage portals
- Golden paths, self-service and paved roads
- AIOps: anomaly detection, AI-assisted incident triage
- LLMOps & MLOps pipelines in a nutshell
- Capstone build, review and presentation

BACKSTAGE

AIOPS

MLOPS

CROSSPLANE

## HANDS - ON

# Six real-time projects

Each project is graded, reviewed by a mentor and lands in your portfolio with a public repository and an architecture write-up.

**P1 Zero-touch CI/CD pipeline**

Build, test, scan and deploy a microservice with GitHub Actions and environment gates.

**P2 Containerised microservices stack**

Multi-stage Docker builds, Compose local stack, hardened and signed images.

**P3 Terraform cloud landing zone**

VPC, IAM, compute and managed database provisioned from reusable modules.

**P4 Kubernetes production deployment**

Helm-packaged app with autoscaling, ingress, secrets and canary rollout.

**P5 GitOps with Argo CD**

Declarative multi-environment delivery, drift repair and one-click rollback.

**P6 Observability & SRE workspace**

Prometheus + Grafana + Loki dashboards, SLOs, alerts and an incident postmortem.

## CAPSTONE

## Ship a production-grade platform, end to end.

Provision cloud infrastructure with Terraform, deliver a containerised application through a secured GitOps pipeline onto Kubernetes, and prove it with SLO dashboards, alerting and a documented incident drill.

TERRAFORM

K8S

ARGO CD

VAULT

GRAFANA



## OUTCOMES

# What you walk away with

### ROLE-READY

DevOps Engineer, SRE, Platform or Cloud Engineer interviews.

### PORTFOLIO

7 documented repositories with architecture diagrams.

### CERTIFIED

Internship, training and course certificates + LOR.

### MENTORED

Live reviews, mock interviews and resume engineering.

## OUR PROCESS

01

### Demo call

See the curriculum and mentors.

02

### Choose a plan

Pick the track that fits your goals.

03

### Enroll

Join the cohort and community.

04

### Engage 2–3 months

Weekly labs, reviews and stand-ups.

05

### Get placed

Portfolio, mock interviews, referrals.

## CERTIFICATIONS



**Internship Certificate**



**Training Certificate**



**Course Certificate**



**Letter of Recommendation**

## WHAT EVERY COHORT INCLUDES

### LIVE MENTOR SESSIONS

Weekly working sessions with practising DevOps engineers.

### LAB ENVIRONMENTS

Guided cloud and cluster labs with reference repositories.

### CODE & DESIGN REVIEWS

Pull-request feedback on every project you submit.

### INTERVIEW PREP

Scenario rounds, troubleshooting drills and resume review.

### LIFETIME RESOURCES

Recordings, runbooks, cheat sheets and community access.

### PLACEMENT SUPPORT

Referrals through our hiring tie-up companies.

# ACCREDITED CERTIFICATIONS.

Every learner receives industry-recognised, accredited certificates on completion of the internship, industrial training and capstone project modules.



**INTERNSHIP CERTIFICATE**  
Accredited Certification



**INDUSTRIAL TRAINING**  
Accredited Certification



**PROJECT COMPLETION**  
Accredited Certification

## INTERNSHIP

Verified internship completion credential

## TRAINING

Industrial training hours certified

## PROJECT

Capstone project completion proof

# GLOBAL MNC CERTIFICATIONS.

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.



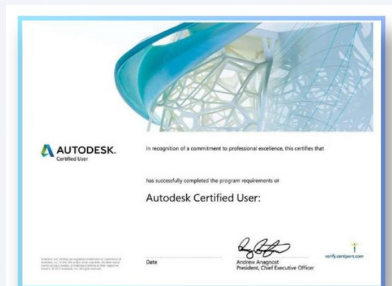
**CISCO**  
Cybersecurity CCST



**MICROSOFT**  
Azure Fundamentals



**IBM**  
Internet of Things



**AUTODESK**  
Certified User



**ESB**  
Entrepreneurship

# AUTHORISED PARTNERS.

Edufyi Tech Solutions is an authorised certification and testing partner for the world's leading technology certification bodies.



ADOBE



MICROSOFT



QUICKBOOKS



AUTODESK



CISCO



ESB

## AUTHORISED TESTING & CERTIFICATION CENTRE

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